

Answer ALL questions.

Write your answers in the spaces provided.

- 1** Plant foods in the diet contain antioxidants, such as vitamin C. Storage time and conditions affect the concentration of vitamin C in these foods.
- (a) Explain how increasing the quantity of plant foods that contain vitamin C in the diet may protect against heart disease.

(3)

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- (5)

(1)

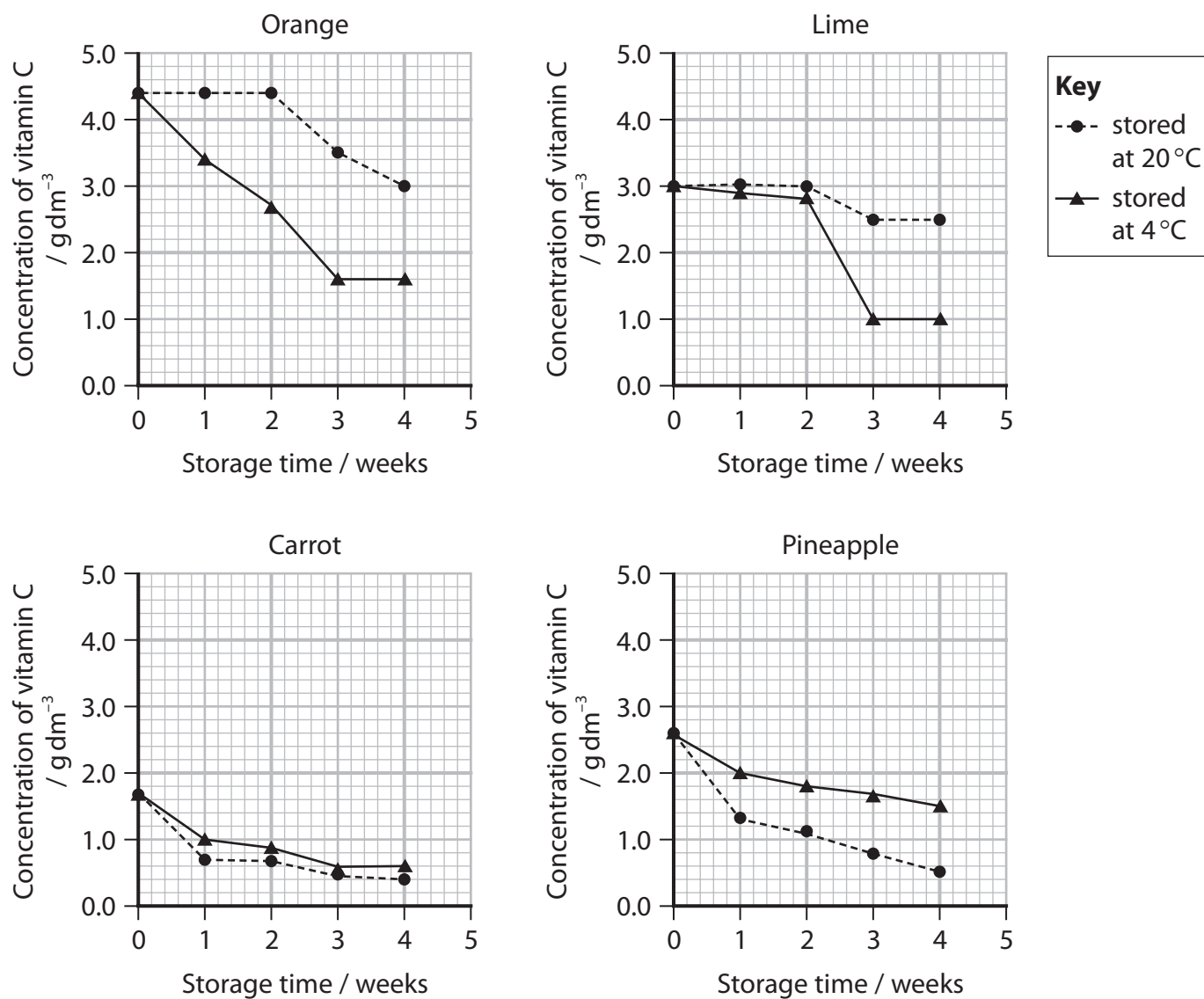


- (c) The effect of storage time and storage temperature on the concentration of vitamin C in different foods was investigated.

The juice from four foods was stored at 4 °C and 20 °C for four weeks.

The vitamin C concentration was measured each week.

The graphs show the results of this investigation.



- (i) Calculate the rate of change of vitamin C content in oranges for the first 3 weeks of storage at 4 °C.

Give your answer to 2 significant figures.

(2)

Answer $\text{g dm}^{-3} \text{ week}^{-1}$

- (ii) Comment on the effects of storage time and temperature on the vitamin C content of these juices.

(4)

(Total for Question 1 = 15 marks)

